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Linear Algebra.

* What is Linear Algebra.

: Linear algebra is the branch of mathematics that studies vector, matrices, and linear transformation.

: It deals with operations such as addition, multiplication, and transformations of data in multiple dimensions.

Algebra : equations with numbers

Linear Algebra : equations with
vectors & matrices

● Vector.

∴ A vector is a quantity that has both magnitude and direction

! It is usually represented as an arrow in a plane or space.

Example.

$$\vec{A} = (3, 4)$$

①

Student Name	Math	physics
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A	62	81
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B	79	73
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C	69	71
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$$\vec{A} = (82, 81)$$

$$\vec{B} = (79, 73)$$

$$\vec{C} = (69, 71)$$

②

subject

New Score:

Math

82

Physics

91

Science

78

Python

95

$$S_2 = (82, 91, 78, 95)$$

- Norm - 1 and Norm 2

Norm - 1

! The L1 Norm of a vector is the sum of absolute values of its components.

! It's called the Manhattan distance because it measures distance along grid-like paths

Example:-

$$\vec{x} = (3, -4, 5)$$

$$\begin{aligned} \|\vec{x}\|_1 &= (3) + (-4) + (5) \\ &= 3 + 4 + 5 = 12 \end{aligned}$$

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Norm - 2

! The L2 Norm of a vector is the square root of the sum of squares of its components.

! This is the usual distance formula.

Example:

$$\vec{x} = (3, -4, 5)$$

$$\|\vec{x}\|_2 = \sqrt{3^2 + (-4)^2 + 5^2}$$

$$= \sqrt{9 + 16 + 25}$$

$$= \sqrt{50}$$

$$\approx 7.07$$

- Dot product and angle between two student's score vector.

• Dot Product.

! The vector dot product is an algebraic operation that takes two equal-length sequences of numbers and returns a single scalar value.

Example.

$$\vec{A} = (3, 4, 5), \vec{B} = (2, -1, 3)$$

$$\begin{aligned}\vec{A} \cdot \vec{B} &= (3)(2) + (4)(-1) + (5)(3) \\ &= 6 - 4 + 15 \\ &= 17\end{aligned}$$

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• Angle between two Vectors:

! The angle between two vectors is the measure of how "aligned" or "different" they are.

Formula.

$$\cos \theta = \frac{\vec{a} \cdot \vec{b}}{\|\vec{a}\| \|\vec{b}\|}$$

$$\text{Let } \vec{a} = (1, 2), \vec{b} = (3, 4)$$

$$\vec{a} \cdot \vec{b} = (1)(3) + (2)(4) = 3 + 8 = 11$$

$$\|\vec{a}\| = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\|\vec{b}\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$\therefore \cos \theta = \frac{11}{\sqrt{5} \cdot 5} = \frac{11}{5\sqrt{5}}$$

• Matrix

! A matrix is a rectangular array of numbers arranged in rows and columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

• Matrix addition.

! Two matrices can be added only if they have the same order (same rows & columns).

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 1+5 & 2+6 \\ 3+7 & 4+8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

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Matrix Multiplication.

! Defined only if columns of A

A = rows of B

Example.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$$

$$AB = \begin{bmatrix} (1 \times 2 + 2 \times 1) & (1 \times 0 + 2 \times 2) \\ (3 \times 2 + 4 \times 1) & (3 \times 0 + 4 \times 2) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 \\ 10 & 8 \end{bmatrix}$$

Linear Transformations & Geometry

- Line, Plane and Hyperplane.

! Our dataset contains four subjects: Math, Science, English History

1. Line

! A line is a one-dimensional geometric object. It has one direction of movement.

In a 2D graph, a line can be represented using an equation such as:

$$y = mx + c$$

2. Plane

! A plane is a two-dimensional flat surface existing in 3D space.

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Think of it like a flat sheet floating inside three-dimensional space.

: If three subjects such as Maths, Science, English are used, each student becomes a point in 3D space. A plane can be used to separate or divide groups of points.

3. Hyperplane

: A hyperplane is the higher-dimensional generalization of a line and plane. In an n -dimensional space, a hyperplane has $n-1$ dimensions.

* Dimensionality increases from $2D \rightarrow 3D \rightarrow$ higher dimensions.

! Dimensionality means the number of independent values to describe a point.

• $2D \rightarrow$ Space $\rightarrow$ Line.

! Using two subjects, Math and Science, each student is represented by a point in $2D$ space. A line can be used as a boundary.

• $3D$ - Space - plane.

! Using three subjects Math, Science and English each student is represented by a point in $3D$ space. A plane can be used as a boundary.

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• 4D or higher - Space - Hyperplane

: Using four subjects each student is a point in 4D space. A 3D hyperplane can be used as a boundary. with our four subject dataset Students are points in 5D space & a 4D hyperplane can be used as a boundary.